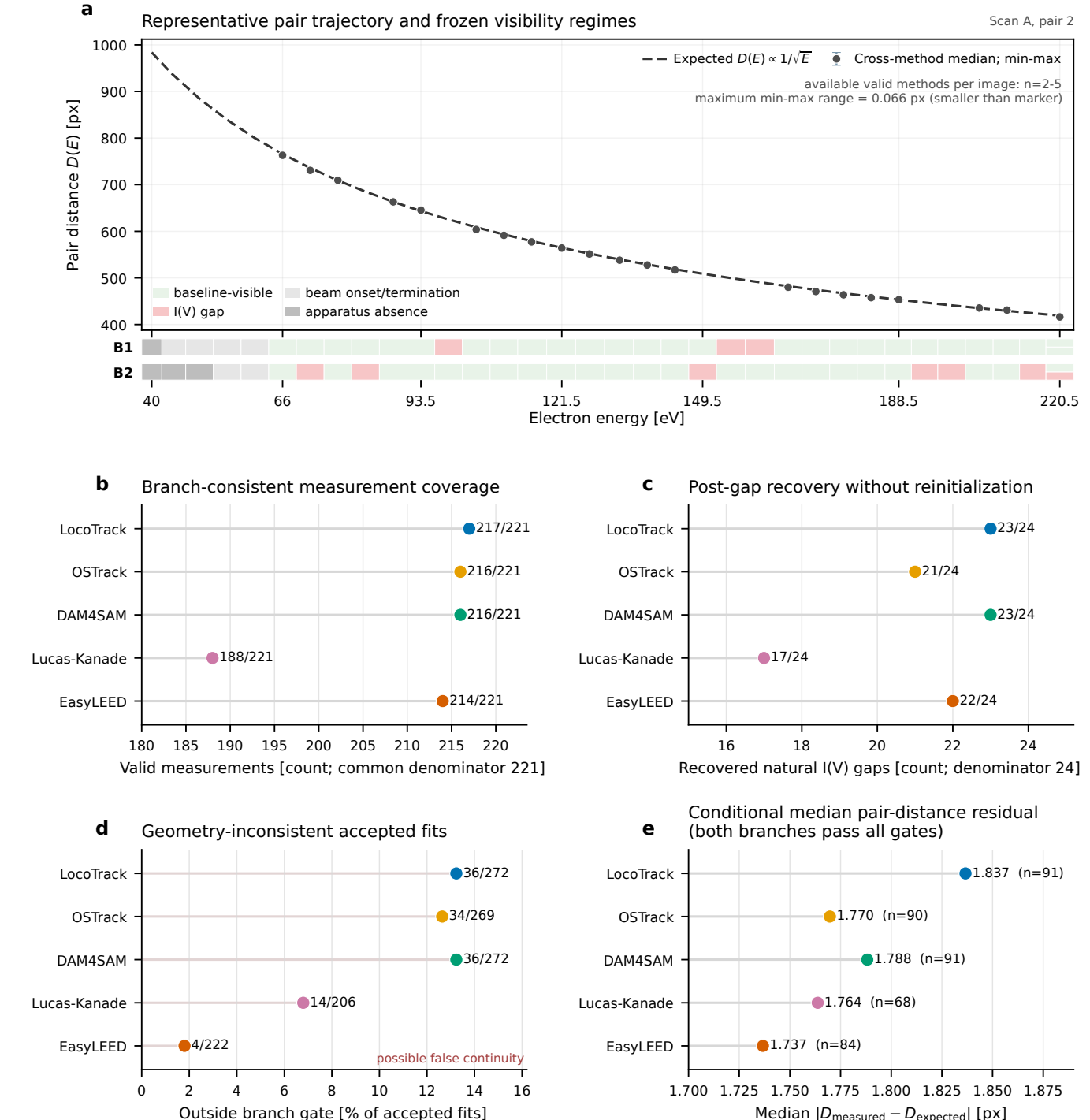
a
Representative pair trajectory and frozen visibility regimes
Scan A, pair 2
Expected $D(E) \propto 1/\sqrt{E}$
Cross-method median; min-max
available valid methods per image: n=2-5
maximum min-max range = 0.066 px (smaller than marker)
Pair distance $D(E)$ [px]
1000
900
800
700
600
500
400
baseline-visible
beam onset/termination
I(V) gap
apparatus absence
B1
B2
40
66
93.5
121.5
149.5
188.5
220.5
Electron energy [eV]
b
Branch-consistent measurement coverage
LocoTrack
OSTrack
DAM4SAM
Lucas-Kanade
EasyLEED
217/221
216/221
216/221
188/221
214/221
180
185
190
195
200
205
210
215
220
Valid measurements [count; common denominator 221]
c
Post-gap recovery without reinitialization
LocoTrack
OSTrack
DAM4SAM
Lucas-Kanade
EasyLEED
23/24
21/24
23/24
17/24
22/24
16
18
20
22
24
Recovered natural I(V) gaps [count; denominator 24]
d
Geometry-inconsistent accepted fits
LocoTrack
OSTrack
DAM4SAM
Lucas-Kanade
EasyLEED
36/272
34/269
36/272
14/206
4/222
possible false continuity
0
2
4
6
8
10
12
14
16
Outside branch gate [% of accepted fits]
e
Conditional median pair-distance residual
(both branches pass all gates)
LocoTrack
OSTrack
DAM4SAM
Lucas-Kanade
EasyLEED
1.837 (n=91)
1.770 (n=90)
1.788 (n=91)
1.764 (n=68)
1.737 (n=84)
1.700
1.725
1.750
1.775
1.800
1.825
1.850
1.875
Median $|D_{measured} - D_{expected}|$ [px]